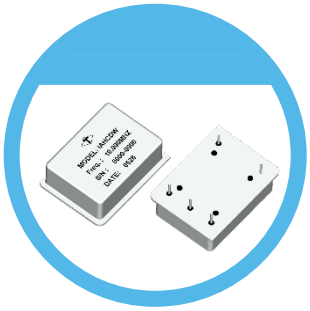
TAITIEN
Our Performance, Your Reputation

NI-100M-6700 Series

100MHz Ultra Low G-Sensitivity Oven Controlled Crystal Oscillator



RoHS Compliant

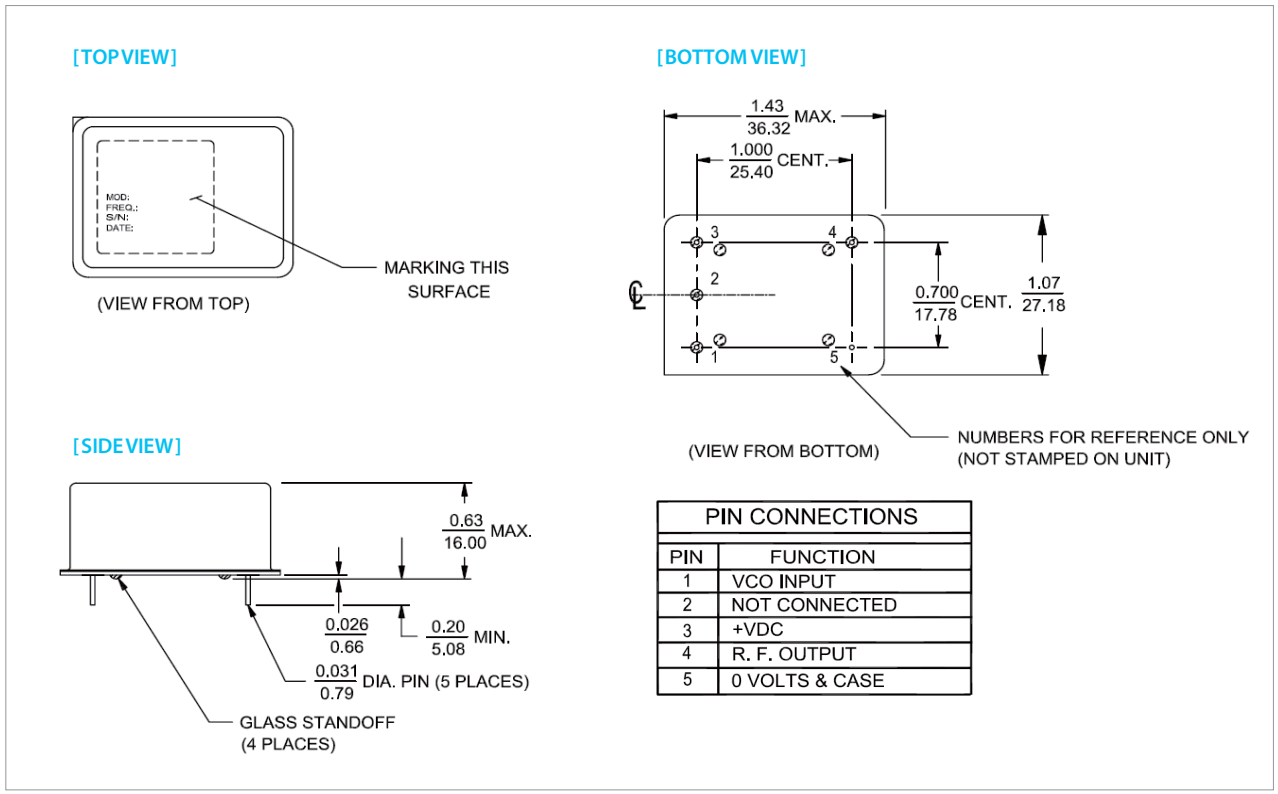
FEATURES

- Ultra Low G-Sensitivity
- Hermetically Sealed Package
- Low Phase Noise
- Tight Frequency Stability
- Fast Warm-up Time
- Electrical Frequency Tuning Input
- Reference Voltage Output
- RoHS-Compliant (lead-free)

TYPICAL APPLICATION

- Instrument Reference
- Microwave Communication
- Clock Reference for Microwave Signal Source
- Test & Measurement
- Telecom Systems
- Radar Systems
- Medical (MRT)

DIMENSION (mm)



PIN CONNECTIONS	
PIN	FUNCTION
1	VCO INPUT
2	NOT CONNECTED
3	+VDC
4	R. F. OUTPUT
5	0 VOLTS & CASE

Note: not all combination of options are available. Other specifications may be available upon request.

Specifications subject to change without notice.

www.taitien.com
sales@taitien.com.tw

ELECTRICAL SPECIFICATION

Test conditions: VDC=+12V; VCO=+5V; at +25 ± 3°C unless otherwise identified

OUTPUT (PIN = "R.F. OUTPUT")

Parameter	Min.	Typ.	Max.	Unit	Test Condition
Frequency (Fo)		100.000000		MHz	
Initial Accuracy	-0.2		+0.2	ppm	after turn on power 30 minutes
					Within 90 days following date code
Waveform		Sine wave			
Level	+7			dBm	
Load		50		Ω	±10%
Harmonics			-30	dBc	
Spurious			-100	dBc	100Hz to 5MHz from carrier

FREQUENCY STABILITY

Parameter		Min.	Typ.	Max.	Unit	Test Condition	
Ambient			±100		ppb	referenced to 25°C	Refer to Table 1 : Ordering Information
			-40°C ~ +85°C		°C		
Aging	Daily	-5		+5	ppb	after 30 days of continuous operation	
	Yearly	-300		+300	ppb		
	10 Years	-1.5		+1.5	ppm		
Voltage		-10		+10	ppb	±5% change	
Load		-5		+5	ppb	±10% change	
Short term				0.1	ppb	root Allan variance for τ =1 sec	
Warm-up		-100		+100	ppb	in 5 minutes	referenced to 1 hour
Phase Noise				-100	dBc/Hz	@ 10Hz	
				-130	dBc/Hz	@ 100Hz	
				-157	dBc/Hz	@ 1KHz	
				-162	dBc/Hz	@ 10KHz	
				-164	dBc/Hz	@ 100KHz	
				-164	dBc/Hz	@ 1MHz	
Phase Jitter(RMS)				0.1	pSec	12KHz ~ 20MHz	
G-Sensitivity (each axis)		Option A	Option B	Option C	Option D		Refer to Table 1 : Ordering Information
		0.5	0.2	0.1	0.05	ppb/g	

ELECTRICAL FREQUENCY ADJUSTMENT (PIN = "VCO INPUT")

Parameter	Min.	Typ.	Max.	Unit	Test Condition
Tuning Range	±2.0			ppm	Referenced to frequency at nominal Center Voltage
Control Voltage	0		+10.0	V	
Slope		Positive			
Center Voltage		+5		V	
Linearity	-10		+10	%	
Modulation Bandwidth	1			KHz	3dB cut off frequency

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INPUT POWER (PIN = "+VDC")

Parameter		Min.	Typ.	Max.	Unit	Test Condition
Voltage		+11.4	+12	+12.6	V	
Current	Steady State			1.9	W	@ +25°C
	During Warm-Up			400	mA	

REFERENCE VOLTAGE (PIN = "REFERENCE VOLTAGE")

Parameter	Min.	Typ.	Max.	Unit	Test Condition
Voltage	+9.5	+10	+10.5	V	

ENVIRONMENTAL

Parameter	Reference Std.	Test Condition
Operable Temperature	-40°C to +85°C	Note 1
Storage Temperature	-55°C to +105°C	
Humidity	MIL-STD-202, Method 103,Test Condition A	95% RH @ +40°C,non-condensing,240 hours
Vibration (non-operating)	MIL-STD-202, Method 201	0.06" Total p-p, 10 to 55 Hz
Shock (non-operating)	MIL-STD-202, Method 213,Test Condition J	30g, 11ms, half-sine

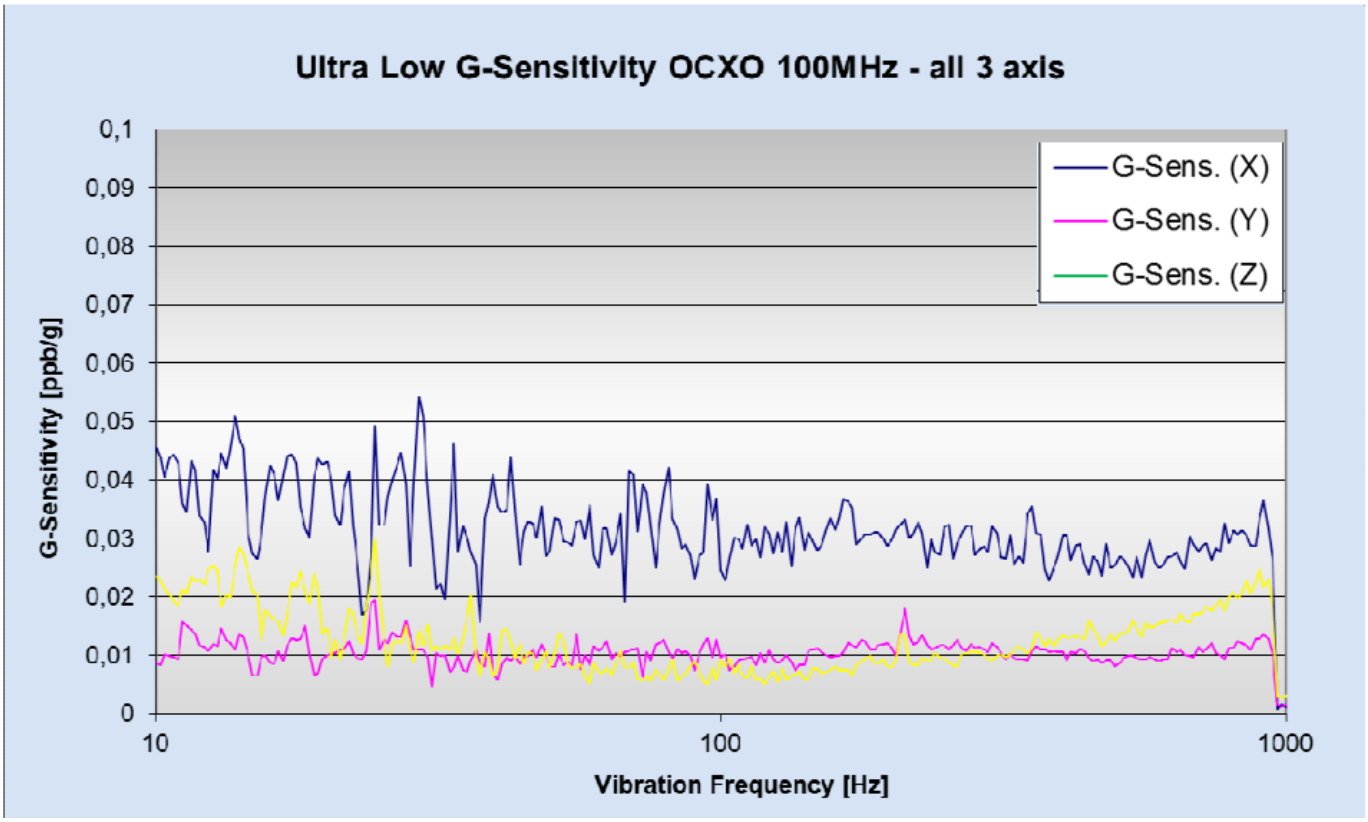
Note 1 : Output maintained over this temperature range. Other requirements of this specification may not be met when operating outside the operating temperature range.

Table 1 : ORDERING INFORMATION

Ambient Temp.(°C) / Option		G-Sensitivity Option			
		A	B	C	D
-40°C~+85°C	± 100 ppb	NI-100M-6700	NI-100M-6701	NI-100M-6702	NI-100M-6703

Other specifications may be available upon request.

G-Sensitivity Test Data



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